## PRODUCTIVITY OF MACHINE

Let us consider a case where an operator works on a drilling machine for 8 hours to make drills on 100 identical work pieces. With the same machine, the operator is able to make drills on 140 work pieces by using jigs and drills with high cutting speeds.

$$\text{Increase in Productivity} = \frac{\text{Change in Production}}{\text{Original Production (Input)}}$$

$$= \frac{140 - 100}{100} \times 100$$

$$= 40\%$$

## INCREASING PRODUCTIVITY METHODS

Some important methods of increasing the productivity are given below,

1. By improving the working conditions.
2. By improving the process involved in production
3. By reducing the non-productivity time by work measurement
4. By providing suitable incentives to the workers.
5. By proposing better plant maintenance programmes.
6. Old and worn out machines should be corrected to make them function as before.
7. Men, Machine and Materials should be maintained at the required quantum.
8. By providing the operators with new and proper training.
9. Layout of the plant and equipments should be improved.

## 11.8 PRODUCTION PLANNING AND CONTROL (PPC)

Production planning is a process of scheming the production procedures to get the finished products of required quality from the raw materials with in a prescribed time frame economically.

Production control is a process of making specific arrangements to carry out the procedures outlined by production planning. It should also over see corrective measures in case of failures in the production process.

## 11.9 FUNCTION OF PPC

1. **Pre-planning:** It involves the decisions of the preparatory functions in the production process.
2. **Route plan:** It is the plan arranging various activities that take place from the stage of raw material to the finished product.
3. **Scheduling:**Scheduling is the preparation of a time table to show the time of starting and the time of completing of operations involved in production of each component.
4. **Despatching**: It is the process of ordering different departments to carry out production process as per the route plan and scheduling,
5. **Controlling**: It is the process of getting reports from all departments regarding production and taking correcting actions, if necessary.

## IMPORTANCE OF PPC

Guideline is very important to all departments. It must be clear and correct.



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